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DEPARTMENT OF STATISTICS

M.Sc. Data Science Application

Statistical Report Project
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The Python code for data manipulation, analysis, and figures is in a private repository for academic integrity, per application rules. Available for
TU Dortmund admissions committee upon request.

Link: <https://github.com/parsafaramarzi/Dortmund-Summer-2026-Application-Statistical-Report>

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1 Introduction

Data-driven decision-making is essential in professional sports. In elite cycling, performance metrics help teams optimize talent identification and resource allocation. This report analyzes rider data [1] to evaluate how specialization, terrain, and outcomes interact. This study examines two questions: Does rider class significantly affect points performance? Does stage class (i.e. flat, hills, mountain) impact points? The analysis tests differences in points across Rider Classes and Stage Classes. Analysis shows significant points differences among Rider Classes. Performance also varies by Stage Class, suggesting rider-terrain interactions. **Section 2** describes the data and variables. **Section 3** details methods, including descriptive and non-parametric tests. **Section 4** evaluates results with visualizations and tests. **Section 5** summarizes and discusses findings.

2 Detailed Description of the Problem

The dataset comes from a cycling manager simulation game by TU Dortmund University [1]. It includes performance metrics for riders across 20 grand tour stages, with 3,496 observations (i.e. one per rider per completed stage). No missing values were found, ensuring reliable analysis. Our analysis uses three variables: dependent variable Points (ratio scale; performance score per rider per stage, with true zero enabling ratios) and independent categorical factors Rider Class (i.e. All Rounder, Sprinter, Climber, Unclassed) and Stage Class (i.e. Flat, Hills, Mountain), both nominal scales [2] for unordered groups. This report investigates rider classes and stage difficulty's influence on points performance, guided by:

Q1: Is there a significant difference in median points among Rider Classes (i.e. All Rounder, Climber, Sprinter, Unclassed)?

Q2: Does points performance vary significantly by Stage Class (i.e. Flat, Hills, Mountain)?

3 Methods

The statistical methodology was structured into three phases—descriptive analysis, assumption checking, and inferential hypothesis testing—to provide evidence-based answers to Q1 and Q2, addressing data complexities like non-normality and heteroscedasticity. Analyses used **Python 3.10** [3], specifically leveraging the **pandas** [4] for data handling, **scipy** [5] for core statistics, **Matplotlib** [6] for plotting, **seaborn** [7] for visualizations and **scikit-posthocs** [8] for post-hoc testing.

3.1 Descriptive Statistics and Visualization

Descriptive statistics were calculated to summarize the central tendency and dispersion of the dependent variable, **Points**, across the nominal categories of **Rider Class** and **Stage Class** [9].

3.1.1 Measures of Central Tendency and Dispersion

For all groups, the **Mean** ($\bar{x}$), **Median** ($\tilde{x}$), and **Standard Deviation** (σ) were calculated to summarize performance [10]. Given the extreme positive skewness and zero-inflation of the Points variable (which are ratio-scaled), the Median was selected as the robust measure of central tendency for all primary comparisons, as it is less sensitive to outliers and non-normal distributions than the mean. The high standard deviation observed across all groups confirmed significant volatility, aligning with the nature of competitive performance data.

3.1.2 Data Visualization

Two visualization techniques provided diagnostic views:

1. **Histograms:** Assessed global distribution, showing right skew and zero-inflation in Points, driving non-parametric tests [11].
2. **Box Plots:** Compared five-number summary (i.e. min, Q1, median, Q3, max), outliers, and IQR spread across groups; confirmed median differences and variance between Rider and Stage Classes [9].

3.2 Assumption Checking

The data was formally tested against the core assumptions required for using parametric tests, such as the Analysis of Variance (ANOVA), to ensure the validity of the hypothesis testing framework [12].

3.2.1 Normality Assessment

Parametric inference requires normal distribution of Points within factor levels. This was violated by the variable's ratio scale and visual diagnostics (see **Section 4.1.1**). , rendering parametric methods inappropriate [13].

3.2.2 Homogeneity of Variance (Levene's Test)

Levene's Test [14] was used to formally test the assumption of Homogeneity of Variance (i.e. Homoscedasticity). The test statistic W is calculated as **Equation 3.1**:

$$W = \frac{(N - k)}{(k - 1)} \frac{\sum_{i=1}^k n_i (\bar{Z}_{i.} - \bar{Z}_{..})^2}{\sum_{i=1}^k \sum_{j=1}^{n_i} (\bar{Z}_{ij} - \bar{Z}_{i.})^2}$$

Equation 3.1: Homogeneity of Variance (Levene's Test)

where Z_{ij} is the absolute deviation of the j^{th} observation from the i^{th} group median. Levene's test showed homogeneity violated for Rider Class but met for Stage Class. However, overall normality

violation required non-parametric methods for all inferential tests, overriding the partial homogeneity success.

3.3 Inferential Statistics

The confirmed and substantial violations of the normality assumption led to the necessary rejection of the parametric ANOVA model. The **Kruskal-Wallis H-Test** was selected as the non-parametric statistical method to test both research questions [15].

3.3.1 Kruskal-Wallis H-Test

Due to the confirmed violation of the Normality and Homogeneity of Variance assumptions, the **Kruskal-Wallis H-Test** [15] was selected as the non-parametric equivalent to one-way ANOVA. This test examines the null hypothesis that the median points are equal across the groups of the independent factors (i.e. Rider Class and Stage Class). The H statistic is calculated based on rank sums according to **Equation 3.2**:

$$H = \frac{12}{N(N+1)} \sum_{i=1}^k n_i (\bar{R}_i - \bar{R})^2$$

Equation 3.2: Kruskal-Wallis H-Test

Where N is the total number of observations, k is the number of groups, n_i is the sample size of group i , $\bar{R}_i$ is the average rank of group i , and $\bar{R}$ is the average rank of all observations.

3.3.2 Dunn's Post-Hoc Test and Bonferroni Correction

A statistically significant Kruskal-Wallis result only indicates a difference exists somewhere among the groups (see **Equation 3.3**). To determine which specific pairs of groups differ significantly, the **Dunn's Post-Hoc Test** [16] was employed. Given the increased risk of Type I error (i.e. false positives) from multiple pairwise comparisons, the conservative Bonferroni correction [17] was applied. This adjustment divides the original significance level ($\alpha_{original} = 0.05$) by the number of comparisons (m):

$$\alpha_{adjusted} = \frac{\alpha_{original}}{m}$$

Equation 3.3: Dunn's Post-Hoc Test

The resulting adjusted p -values are reported in **Section 4.2**.

4 Evaluation

The evaluation phase synthesizes the descriptive analysis and inferential hypothesis testing to provide robust answers to the research questions. The results demonstrate that rider performance is not random but highly structured by specialization and terrain.

4.1 Descriptive Analysis

Descriptive statistics reveal a performance landscape characterized by extreme skewness, zero-inflation, and clear hierarchical dominance.

4.1.1 Global Data Distribution (Normality Check)

Analysis examined total points distribution for 184 riders across the tour.

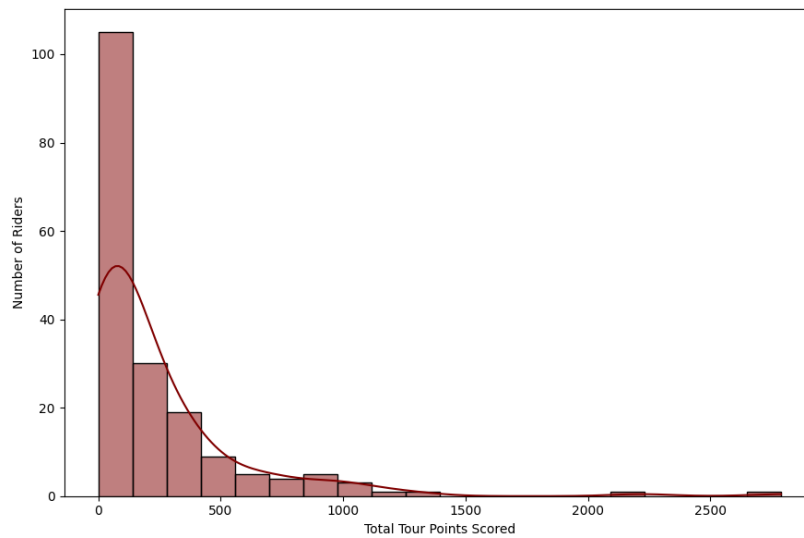


Figure 4.1: Histogram of Total Tour Points

Figure 4.1 shows severe right-skewness and zero-inflation: most observations near zero (i.e. minimal points for majority), long tail for elite accumulators. This violates normality, necessitating non-parametric tests [18] [13].

4.1.2 Rider Class Hierarchy (Addressing Q1)

Figure 4.2 clearly establishes the performance hierarchy: All-Rounders demonstrate the highest median total score, confirming their primary contender status. The observed hierarchy is **All-Rounder > Climber > Sprinter > Unclassed**, with the latter group showing minimal point acquisition, confirming scarcity for non-specialists.

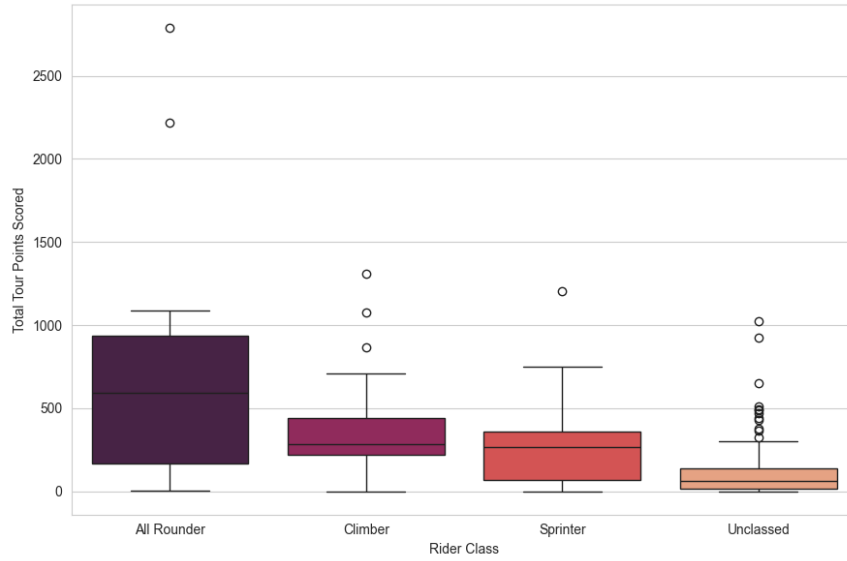


Figure 4.2: Box Plot of Total Tour Points by Rider Class

4.1.3 Stage Class Interaction (Addressing Q2)

The interaction between rider type and terrain difficulty is visualized in **Figure 4.3**, with numerical foundations detailed in **Table 4.1**.

Table 4.1: Descriptive Statistics by Rider Class and Stage Class

Rider Class	Stage Class	Count	Mean	Median	STD	min	max
All Rounder	flat	102	15.44	8.00	28.28	0	194
All Rounder	hills	136	35.78	12.50	57.45	0	274
All Rounder	mount	85	67.42	17.00	88.95	0	304
Climber	flat	138	5.09	1.50	6.23	0	26
Climber	hills	184	21.66	7.00	45.98	0	264
Climber	mount	115	35.86	12.00	57.02	0	269
Sprinter	flat	174	38.97	2.50	63.58	0	272
Sprinter	hills	232	5.20	0.00	21.95	0	244
Sprinter	mount	145	2.04	0.00	5.88	0	32
Unclassed	flat	690	5.74	0.00	19.80	0	232
Unclassed	hills	920	9.09	0.00	30.66	0	260
Unclassed	mount	575	2.95	0.00	7.90	0	64

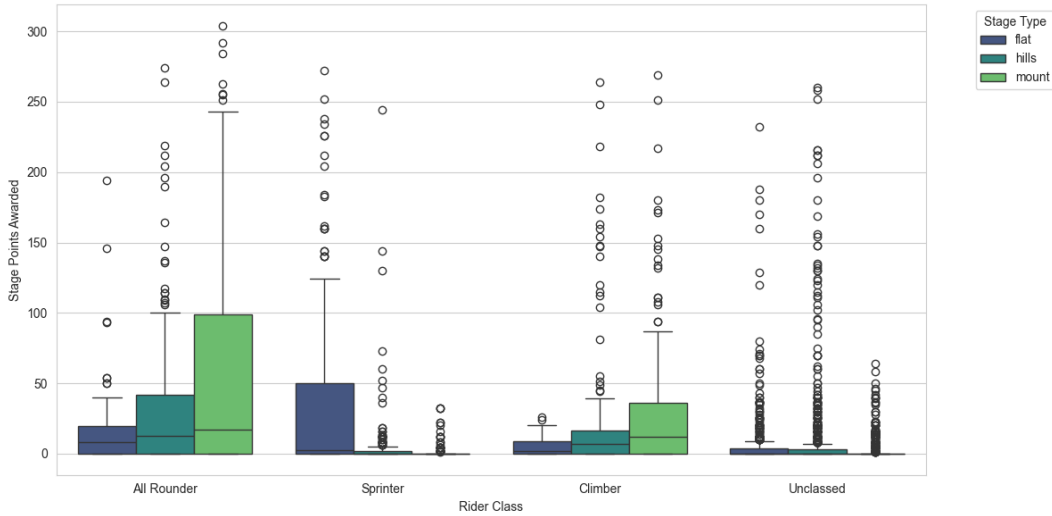


Figure 4.3: Comparative Box Plot of Points by Rider and Stage Class

The data shows clear specialization: Sprinters excel on Flat stages ($\bar{x} \approx 38.98$) but their performance collapses on Mountain stages ($\bar{x} \approx 2.04$), where the median score is effectively zero. Conversely, All Rounders demonstrate superior versatility, peaking on Mountain stages ($\bar{x} \approx 67.42$), achieving the highest average score of any group on any terrain. Variability is high for All-Rounders on Mountain stages, as confirmed by the extensive whiskers in **Figure 4.1**, showing a high variance ($\sigma \approx 88.96$). This indicates success is driven by a few elite performances, not consistent average scoring.

4.2 Hypothesis Testing and Inferential Results

Based on the confirmed non-normality (**Figure 4.1**) and the unequal variances for Rider Class (i.e. Levene's Test: $W=83.08$, $p < 0.001$), the non-parametric **Kruskal-Wallis H-Test** was employed.

4.2.1 Assumption Check: Levene's Test

Levene's Test was conducted to formally check the Homogeneity of Variance assumption:

- **Rider Class:** $W= 83.0757$, $p \approx 0.0000$. The null hypothesis was rejected, confirming significantly unequal variances.
- **Stage Class:** $W= 0.2414$, $p \approx 0.7855$. The null hypothesis was not rejected, indicating statistically equal variances.

Despite the homogeneity for Stage Class, the global non-normality mandated the consistent use of the Kruskal-Wallis test for both factors.

4.2.2 Q1: Rider Class Performance

The Kruskal-Wallis test yielded a highly significant result ($H = 330.20$, $p < 0.001$), confirming a difference in median performance between the classes. Dunn's Post-Hoc Test (summarized in **Table 4.2**) revealed that All Rounders are statistically superior to all other classes ($p < 0.001$), and Unclassed riders are statistically inferior to all specialized classes. Importantly, the difference between Sprinters and Climbers was not statistically significant ($p = 0.0656$), suggesting they occupy a similar middle tier of overall median performance.

Table 4.2: Dunn's Post-Hoc P-value Matrix (Rider Class)

	All Rounder	Climber	Sprinter	Unclassed
All Rounder	1.0000	0.0656	0.0000	0.0000
Climber	0.0656	1.0000	0.0000	0.0000
Sprinter	0.0000	0.0000	1.0000	0.0268
Unclassed	0.0000	0.0000	0.0268	1.0000

4.2.3 Q2: Stage Class Interaction

The Kruskal-Wallis test for Stage Class also yielded a significant result ($H = 9.49$, $p = 0.0086$), confirming that terrain type influences scoring distribution (see **Table 4.3**).

Dunn's Post-Hoc Test clarified this finding:

Table 4.3: Dunn's Post-Hoc P-value Matrix (Stage Class)

	flat	hills	mount
flat	1.0000	0.7300	0.0068
hills	0.7300	1.0000	0.0981
mount	0.0068	0.0981	1.0000

- The significant difference is driven entirely by the comparison between **Flat** and **Mountain** stages ($p = 0.0068$).
- **Hills** stages act as a statistical middle ground, showing no significant median difference from either Flat ($p = 0.73$) or Mountain ($p = 0.09$) stages. This suggests that the impact of terrain on scoring is most pronounced only at the extremes of the difficulty spectrum.

5 Summary and Discussion

Analysis using Kruskal-Wallis H-Test and Dunn's post-hoc confirmed two key findings: 1) Rider Class significantly influences performance ($H=330.20$, $p < 0.001$), with All-Rounders demonstrating statistical superiority over all other classes, validating the premium placed on versatility in multi-stage tours. Climbers and Sprinters were statistically indistinguishable in overall median performance. 2) Stage Class significantly influences scoring ($H=9.49$, $p=0.0086$), driven entirely by the contrast between Flat and Mountain stages. This mandates that team strategy prioritizes Mountain stages for statistical separation. Future work should use multi-level models for nested data.

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